Linker (LinkCreator) — User Guide

Linker is a Niagara Program Object that makes, checks or removes links in bulk: one source to many targets, many sources to one target, or any list of pairs from a CSV. Every link is checked after it's made, and Reverse undoes the last run.

The program calls itself **LinkCreator** in its logs and quickGuide. Covers **N4 v3.3 / v3.4** and **AX v3.3-AX** (Niagara AX 3.5–3.8); see AX notes.

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1. How it works in one minute

```
linkSource (one "ord,slot" per line)  ──►  Direct: targetOrd . targetSlot      (many → one)
                                      ──►  BQL:    each query row . targetSlot (many → many)
CSV file of pairs                     ──►  CSV:    exactly the pairs listed
```

- Before a link is made, both slots must exist. After, it's read back; a link Niagara dropped or can't activate (wrong slot type, unresolved source, licence limit) is removed again and reported as `ERROR`, so you never end up with dead links.
- A link that already exists is skipped (no duplicates).
- Links are named `link_<source>_<slot>_to_<targetSlot>_<number>`. Component Copier and ForceRemove use the same names, so the three tools recognise each other's links.
- It only runs when you press an action. Dry Run and Verify make no changes.

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2. Every field, explained

Legend: **Set** = you set it · **Out** = the program writes it (read only). The display name (what the property sheet shows) is in brackets where it differs.

About

Field		What it is
`version`	Out	The program version.
`quickGuide`	Out	Short built-in help.

Last run

Field		What it is
`status`	Out	Live progress, then the summary (linked, skipped, errors).

What to link

Field		Default	What it is
`operationMode`	Set	BQL	**Direct / BQL / CSV**. See the three modes.
`sourceNote` (Note:)	Out		A fixed reminder of the `linkSource` format, typed in once when the program is built. The code never changes it.
`linkSource` (Source)	Set	blank	Direct and BQL: the source(s), one `ord,slot` per line, e.g. `slot:/Test/BXXXXX_OAT,out`. No slot (just the ord) = `out`. `#` lines are ignored. Not used in CSV mode.
`targetOrd` (Target)	Set	blank	Direct: the target component. BQL: a query, each row a target. CSV: the CSV file of pairs.
`targetSlot` (Target Slot)	Set	blank	Direct and BQL: the slot on the target the link writes to, e.g. `in10`.

Names can be typed as Workbench shows them (`VAV X-01` is escaped for you).

Files and logs

Field		Default (blank =)	What it is
`logFilePath`	Set	`file:^logs/LinkCreator.log`	Step-by-step log.
`resultsCsvPath`	Set	`file:^logs/LinkCreator_results.csv`	One row per link.

Field		Default (blank =)	What it is
`sampleCsvPath`	Set	`file:^logs/LinkCreator_SAMPLE.csv`	Where **Create Sample CSV** writes a starter CSV.
`maxArchives` (Max Archives to Keep)	Set	10	Old copies of the log / CSV kept (0–30; 0 = keep all).

All paths must be inside the station folder (`file:^…`).

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3. The three modes

Direct — many sources → one target

```
linkSource:   slot:/Drivers/BacnetNetwork/BXXXXX_AHU_1/points/OAT,out
targetOrd:    station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101/points/Logic
targetSlot:   in10
```

Several source lines all link to the same `targetSlot`, so normally use one line in Direct mode unless the target slot accepts several links.

BQL — sources → every target the query finds

```
linkSource:   slot:/Drivers/BacnetNetwork/BXXXXX_AHU_1/points/OAT,out
targetOrd:    station:|slot:/Drivers|bql:select * from control:NumericWritable where name = 'OutsideAir'
targetSlot:   in10
```

Every `OutsideAir` point gets a link from the OAT into `in10`. Test the query in Workbench first to see how many rows it returns.

CSV — exactly the pairs you list

`targetOrd` = `file:^Csv/Links.csv`. Five columns, header row required:

```
BOrd1,Slot1,Direction,BOrd2,Slot2
station:|slot:/Drivers/BacnetNetwork/BXXXXX_AHU_1/points/OAT,out,>,station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101/points/Logic,in10
station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_102/points/Logic,out,
<,station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_102/points/DamperCmd,in16
```

- **Direction** `>` = from 1 to 2 · `<` = from 2 to 1.
- A blank Slot1 / Slot2 = `out`.
- `linkSource` and `targetSlot` are ignored in CSV mode.

- Bad rows are listed in the results CSV: `VALIDATION` = caught before the run (bad ord, missing column), `ERROR` = failed while linking.

4. Actions

Action (slot name)	What it does
Execute (`execute`)	Make the links.
Dry Run - No Changes (`dryRun`)	Preview: every link listed as `DRYRUN`, with the reason if it would be skipped.
Verify - Audit (`verify`)	Checks each link the settings describe: `FOUND`, `MISSING`, or `BROKEN` (there but not working). No changes.
Reverse Changes (`reverse`)	Removes exactly the links the last Execute made, newest first. A link changed since is skipped. Not a preview.
Cancel Request (`cancel`)	Stop after the current row.
Create Sample CSV (`createSampleCsv`)	Write a starter 5-column CSV to `sampleCsvPath`.
Remove Old Logs (`pruneArchives`)	Apply `maxArchives` now.

5. Files it writes

File (in `Logs`)	Contents
`LinkCreator.log`	Step by step.
`LinkCreator_results.csv`	Timestamp, SourceName, SourceSlotPath, TargetName, TargetSlotPath, Status, Message, Mode, LinkName, DurationMs.
`LinkCreator_<program name>_reverse.csv`	The links the last Execute made (used by Reverse).
`…_<date>_<time>.log / .csv`	Archives of earlier runs.

6. Complete setups

Placeholders throughout.

A. Outside air temperature to every VAV's logic (BQL)

Field	Value
operationMode	BQL
linkSource	`slot:/Drivers/BacnetNetwork/BXXXXX_AHU_1/points/OAT,out`
targetOrd	`station:|slot:/Drivers/BacnetNetwork|bql:select * from control:NumericWritable where name = 'OutsideAir'`
targetSlot	`in10`

Dry Run → Execute → Verify.

B. One link (Direct)

Field	Value
operationMode	Direct
linkSource	`slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101/points/ZoneTemp`
targetOrd	`station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101/points/Logic`
targetSlot	`in10`

(No `,slot` on the source = `out`.)

C. A spreadsheet of links (CSV)

1. **Create Sample CSV** → open `logs/LinkCreator_SAMPLE.csv` in Excel.
2. Fill in one row per link; save to `Csv/Links.csv` in the station's Files.
3. `operationMode` = CSV, `targetOrd` = `file:^Csv/Links.csv`.
4. Dry Run → fix any `VALIDATION` rows → Execute → Verify.

D. Check links after a change

Leave the settings as they were for the run and press **Verify**. Every link should come back `FOUND`; `BROKEN` ones exist but don't work (check the slots and the source).

E. Undo

Press **Reverse** before the next Execute.

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7. Results CSV

Status	Meaning
`LINKED`	Link made and working.

Status	Meaning
`DRYRUN`	Dry Run: would be made (or skipped — see Message).
`SKIPPED`	Not made; Message says why (already exists, ...).
`FOUND` / `MISSING` / `BROKEN`	Verify results.
`REMOVED` / `DELETED`	Reverse took it away.
`VALIDATION`	CSV row rejected before the run.
`ERROR`	Failed (or made but not working, and removed again); Message has the reason.

8. Good habits

- Back up the station. Dry Run, then Execute, then Verify.
- In BQL mode check the query's row count first.
- Reverse straight away if something went wrong (last run only).

AX notes

- Program name **Linker4AX**, code `Linker4AX_v3.3.java`; AX 3.5–3.8.
- Same slots and actions; no imports to add. `targetOrd` is shown as "Target, BQL or CSV".
- The AX version doesn't sort its own property sheet (that's N4 v3.4).